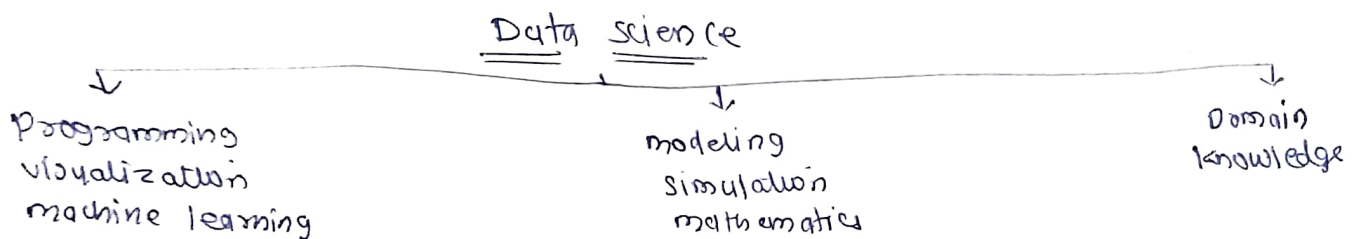


Data science

multiple disciplinary science with an objective to perform data analysis to generate knowledge that can be used for decision making

A data science application collect data and information from multiple heterogeneous sources, clean, integrates, processes and analyses this data using various tools and presents information and knowledge in various visual forms



Types of data

- 1) structured data - RDBMS
- 2) semi structured data - use of tags XML
- 3) unstructured data → large & written textual
- 4) Data streams - characterised by a sequence of data over a period of time.

Descriptive method

used to present basic summaries about data, however, it make no attempt to interpret the data, include statistical values & certain graphs

Exploratory Analysis

as a group of methods that can be used to learn possibilities of relationship among data

Predictive Analysis

uses large amount of data to identify potential risk and aids the decision making process.

Application of data science

web searching, Healthcare system, transport sector.

Data science cycle

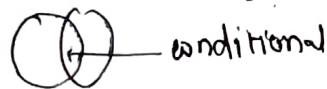
- 1) Data science project requirement Analysis phase - Identify the objectives of data science project
- 2) data collection and preparation phase - checked for ~~dup~~ duplication of data, consistency of data, missing data, and availability of data
- 3) Descriptive data analysis → generate descriptive information about the data.
- 4) Data modeling and model testing → All data models are then tested for their validity with ~~the~~ test data.
- 5) model deployment and Refinement →

PROBABILITY

Probability is a possible measures of Occurrence of a specific event amongst a group of events, if the occurrence of the events is observed for a large number of trials.

conditional probability

defined for the probability of occurrence of an event, when another event has occurred.



Bayes Theorem

deals with the conditional probability

$$P(X/Y) = \frac{P[Y/X] \times P[X]}{P[Y]}$$

Data quality

completeness, Accuracy, timeliness, consistency, validity, unique_{nn}

Data Preprocessing

↓
data cleaning → missing values, manually enter the omitted value
↓
fill up the blank with a global constant, noisy data
data integration entity identification problem, redundancy and correlation analysis, Tuple Duplication
↓
data transformation → Num. of record, attributes or dimension can be reduced, dimensionally reduction, numerosity,
↓
Data Reduction → used to changed the data into formats that are suited for the analytical process

Data ~~creat~~ curation

creating, organising and managing data sets so that people looking for information can access and use them.

Steps:-

Data integration

- * Creates coherent data storage by combining data from several sources.
- * crucial because it maintains data accuracy while providing a consistent view of dispersed data.

Box plots

It is a visual depiction of data that aids in determining how widely distributed or how much the data values change. These boxplots make it simple to compare the distribution since it takes the centre, ~~the~~ spread and overall range understandable.

Example:- Gas efficiency of vehicles

Time spent reading across readers.

Scatter plots

most commonly used chart when observing the relationship between two quantitative variables. It works particularly well for quickly identifying possible correlation b/w different data points.

Fig.

Big data

refers to a collection of information that is not only extremely large in quantity but also growing at an exponential rate over the course of time.

* characteristics of big data

1) volume, velocity, variety, veracity (accuracy), value, validity, volatility, variability, visualization

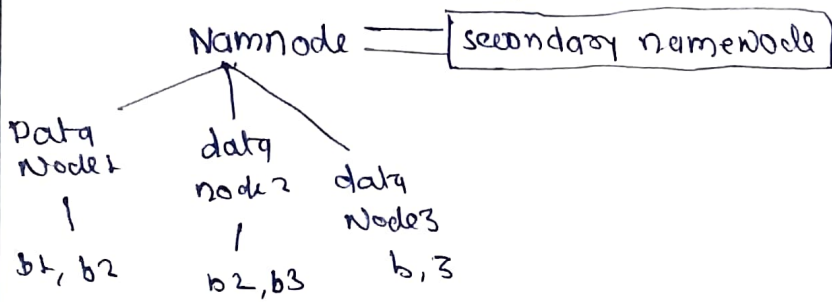
* Application

social media, financial and banking, healthcare, business telecommunication & multimedia, government

Hadoop Distributed file system

collection of several software services that are all freely accessible to the public and can be used in conjunction with one another.

It offers a software framework for storing a huge amount of data in a variety of location and for working with that data by utilising the mapReduce programming ~~style~~ style



HADOOP1

mapReduce
(Resource management +
Data processing)

HDFS

HADOOP2

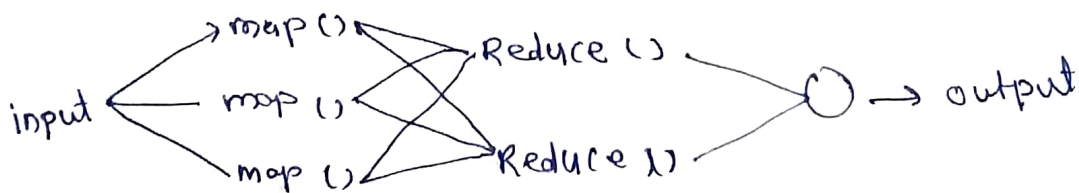
mapReduce +
other Types of job

YARN
Resource management

HDFS

MAP Reduce operation

Parallel processing, data locality, fault tolerant manner



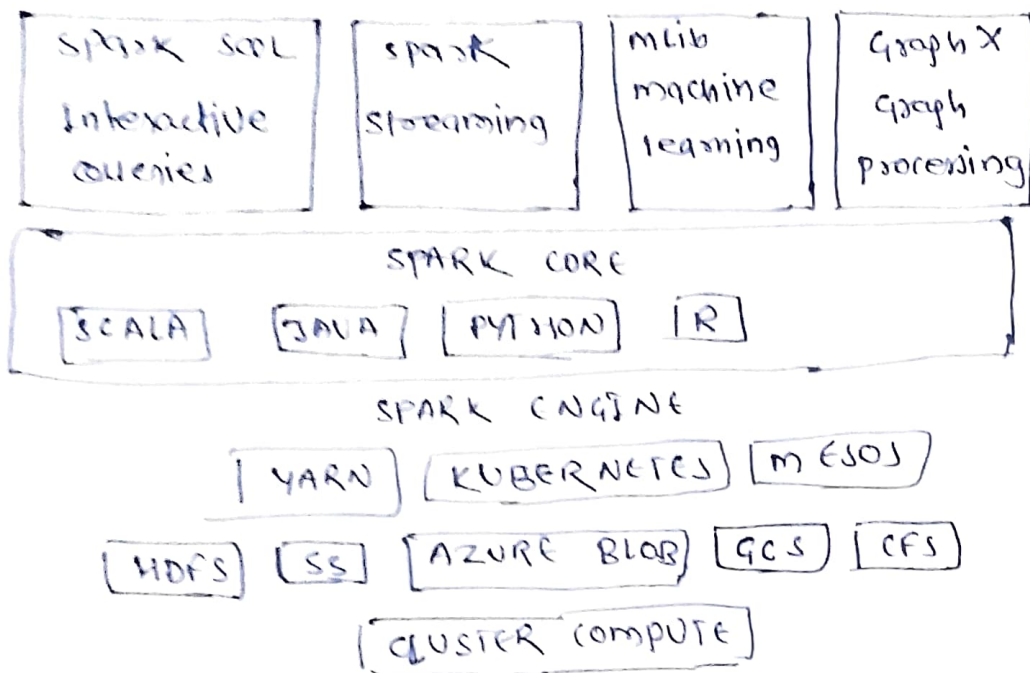
mapReduce architecture

Apache spark framework

Spark is a data processing framework, which can swiftly conduct operations on huge dataset and distribute operations across several computers, either alone as well as in conjunction with other distributed computing tools.

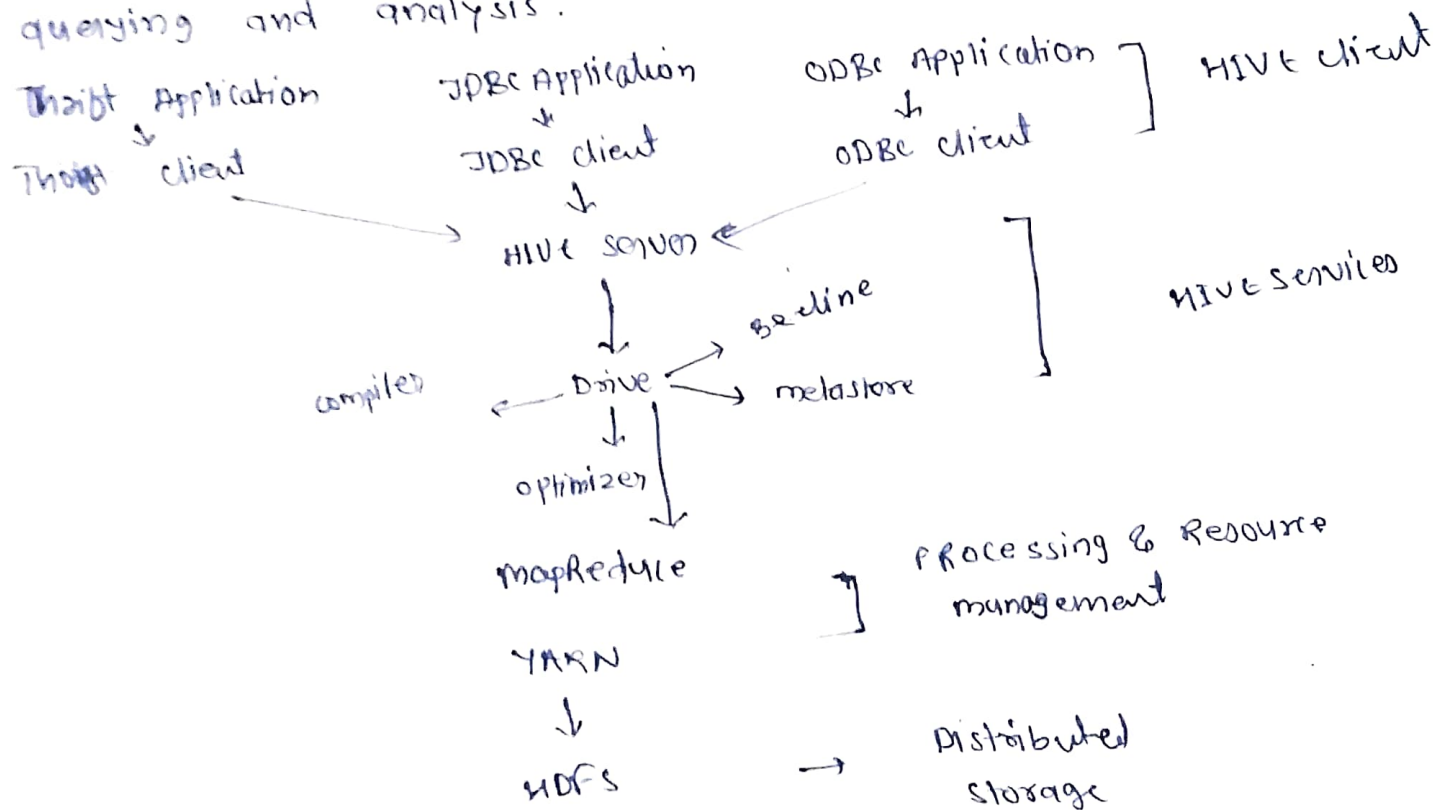
1) Fast processing, multilanguage support, Advance Analytics support.

The main component of Apache Spark



HIVE

Hadoop utility for processing structured data is called Hive. It sits on top of Hadoop to summarize big data and simplifies querying and analysis.



HBase

scales linearly, support automatic failure, offer reliable reading and writing. It is a both source and a destination of Hadoop integration. data replication

Master

Region Server
memstore

HDFS

HFile

write

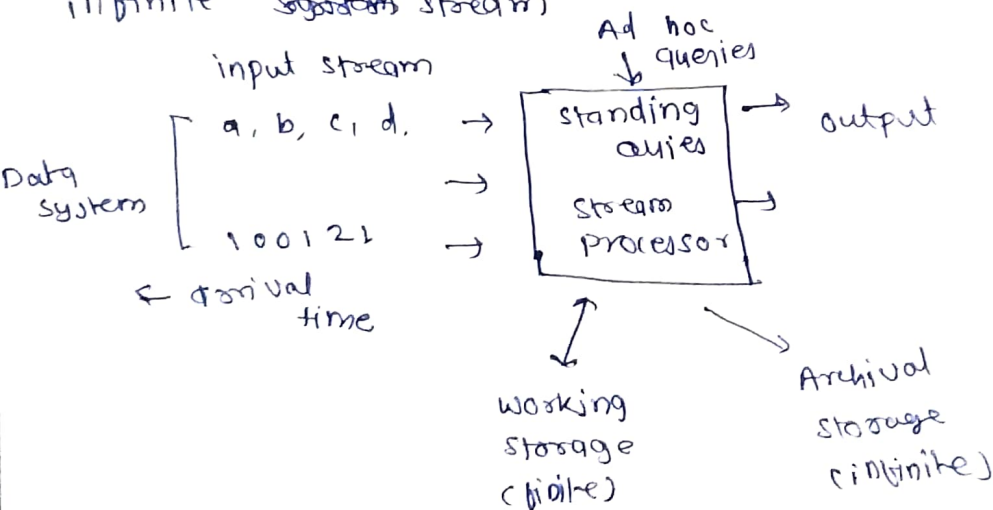
Ahead log

zookeepers

/hbase/region1
:
/hbase/region

Data Streams

mining data streams is a process to extract knowledge in real time from a large amount of volatile data, which comes in an infinite stream



Bloom filters

Bloom filters is used for cases where the number of items on the list or set is so large, that you cannot do a comparison of each stream element with element of the list or check the set membership.

Page Ranking

is an algorithm that was designed for the initial implementation of google search engine. The PageRank is defined as an approach for computing the importance of the web pages in a big web graph.

ENVIRONMENT OF R

designed for statistical analysis of data. It also has a very good support for graphical representation of data. It has a vast set of commands.

Data cleaning and pre processing

is the process of identifying, correcting, and removing incorrect raw data.

Chi-Square test

Statistical tool for determining if two categorical variables are significantly correlated.

Linear regression

Regression analysis is a common statistical technique for establishing a relationship model between two variables.

$$Y = a + bX$$

Simple linear regression

Multiple linear regression $Y = a + b_1X_1 + b_2X_2 + \dots + b_nX_n$

Logistic regression

is a classification algorithm for determining the probability of event success and failure.

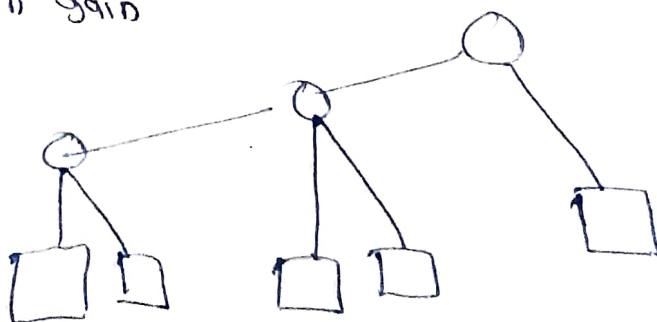
$$g(z) = \frac{1}{1 + e^{-(a + b_1x_1 + b_2x_2 + \dots)}}$$

Decision tree

is a graph that represents decisions and their results in a tree format. Graph nodes represent events or selections and graph edges represent decision rules or conditions.

1) ENTROPY

2) Information gain



Random forest

are a set of decision tree that are used in supervised learning algorithms for classification and regression analysis but primarily for classification.

randomforest (formula, data)

classification

Predict the target class by analysing the training dataset use the training dataset to get better boundary condition that you can use to determine each target class once the constraints are determined, the next task is to predict the target class.

classifier, characteristics, Binary classification, multi class classification, multi label classification

clustering

is an unsupervised learning technique that divides a dataset into multiple groups and is called a cluster because of its similarities.

* clustering method

hard clustering, soft clustering

⇒ k means (x, centres, nstart)

Association rule

unsupervised non linear algorithm to discover how any item is associated with other.

Different mechanisms of finding pageRank

1) simple recursive formulation technique

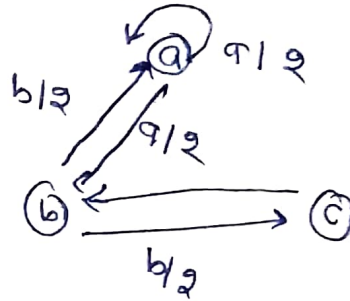
Each site is considered as a vote and the importance of a given vote is proportional to the importance of the source web page that is casting this vote or that is creating a link to the destination

$$r_j = r_j / 2 + r_k / 3$$

1> The flow method

basically the ~~val~~ flow through the network

$$x_j = \sum_{i=j} \frac{r_i}{d_i}$$



Data Structure in R

Vector - one dimensional array of data element that have same data type. logical, integer, double, complex, character data types.

String - Any value written within a pair of single quotes or double quotes in R is treated as a string

> paste('one', 2, 'three', 4, sep = "&")
[1] "one & 2 & three & 4"

- * Describe data science, it uses descriptive, exploratory and predictive
- * Statistical hypothesis testing, error in hypothesis testing
- * Data processing, quality measures in data processing,
- * strategies for data handling
- * Big data
- * Hadoop architecture
- * map reduce
- * Apache spark, HIVE, HBase
- * column database, Document database, graph database
- * Data stream bloom ~~filter~~ filter
- * page rank, mechanisms of finding PageRank
- * Data structure in R
- * classification, clustering, Association rule
- * Random forest Algorithm
- # Statistical hypothesis testing

MCS 226 Data science and Big data

- 1) Data science, Application, categorical and quantitative data
- 2) Big data, characteristics of Big data
- 3) Map reduce paradigm, example
- 4) Similarity, Jaccard similarity of set, example
- 5) Data stream, Now data stream processing different from transaction processing
- 6) Program using R to add and subtract two matrix
- 7) Classification, step about how R Programming can be used to perform classification
- 8) Measures for defining the spread of a quantitative variable? explain
- 9) sampling distribution, example
- 10) Concept of data pre-processing and data cleaning, example
- 11) box plot, sample box plot draw and explain
- 12) Characteristics of HDFS, Advantages over of HDFS over file allocation table (FAT) based system
- 13) Role of shuffling and sorting in map reduce - example
- 15) Features of SPARK architecture explain
- 16) NoSQL database, O/R b/w relational database system
- 17) collaborative filtering, example
- 18) Bloom filtering example
- 19) Page Ranking algorithm example
- 20) issue related to mining of social network
- 21) Define terms in context of R programming
 - i) complex data type
 - ii) %>% operator
 - iii) <- or <-> operator
 - iv) vectors
 - v) strings

- 22) use of box chart, how to draw box chart using R programming
- 23) clustering, steps of how R progr. can used to perform K mean clustering, what is confusion matrix in these context